Phase 2: Data Audit & Forensics

March 31, 2026

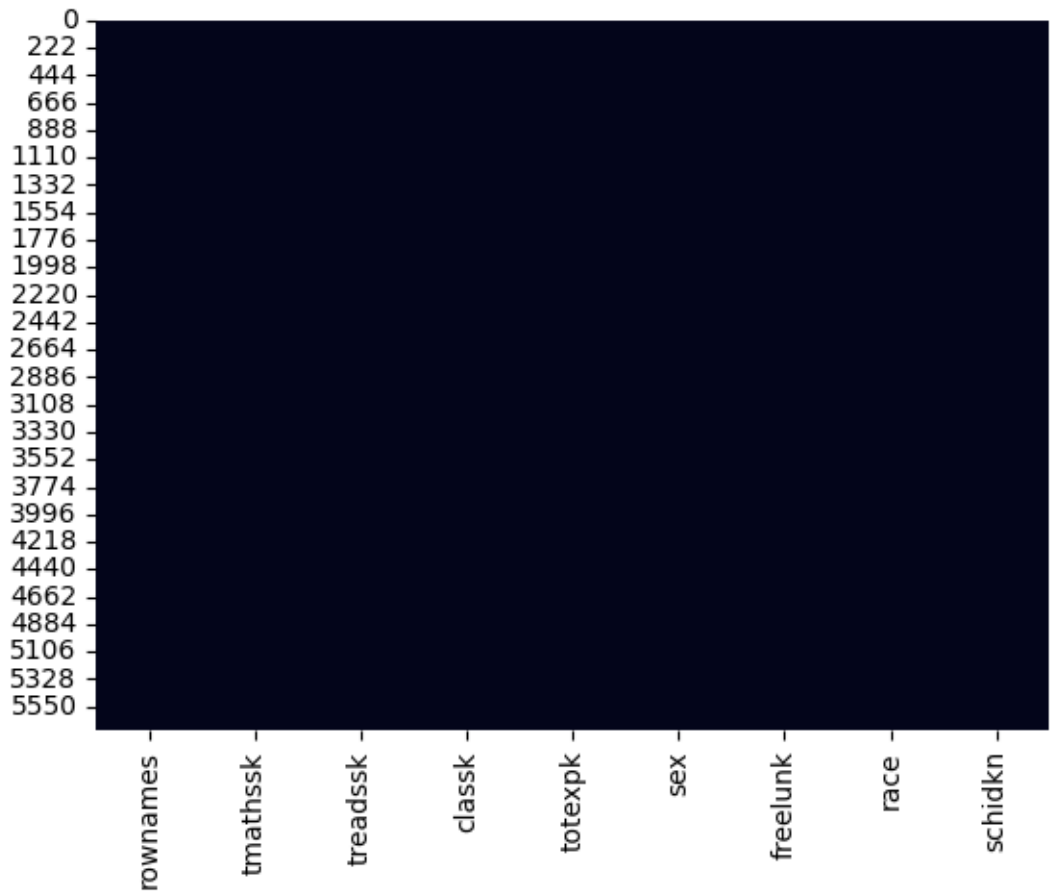
[50]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
```

[51]:
```
df = pd.read_csv('star.csv')

print(df.shape[0])
print(df.shape[1])

print(df.info())
print(df.describe())

df['totexpk'] = df['totexpk'].astype(str).str.replace('$', '').str.replace(',',␣
↪'').astype(float)
sns.heatmap(df.isnull(), cbar=False)
```

```
5748
9
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5748 entries, 0 to 5747
Data columns (total 9 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   rownames  5748 non-null   int64
 1   tmathssk  5748 non-null   int64
 2   treadssk  5748 non-null   int64
 3   classk    5748 non-null   object
 4   totexpk   5748 non-null   int64
 5   sex       5748 non-null   object
 6   freelunk  5748 non-null   object
 7   race      5748 non-null   object
 8   schidkn   5748 non-null   int64
dtypes: int64(5), object(4)
memory usage: 404.3+ KB
None
           rownames     tmathssk     treadssk      totexpk      schidkn
```

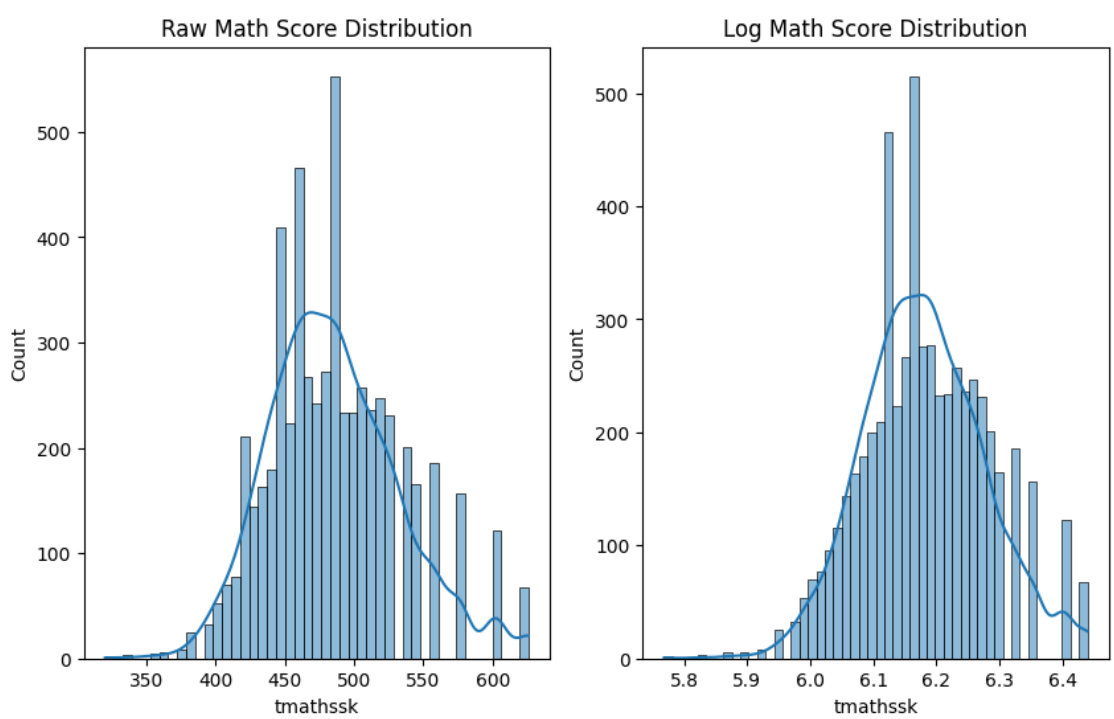
```
count   5748.000000  5748.000000  5748.000000  5748.000000  5748.000000
mean    5752.660404   485.648051   436.742345     9.307411    39.836639
std     3329.971901    47.771531    31.772857     5.767700    22.957552
min        2.000000   320.000000   315.000000     0.000000     1.000000
25%     2884.750000   454.000000   414.000000     5.000000    20.000000
50%     5733.500000   484.000000   433.000000     9.000000    39.000000
75%     8624.250000   513.000000   453.000000    13.000000    60.000000
max    11598.000000   626.000000   627.000000    27.000000    80.000000
```

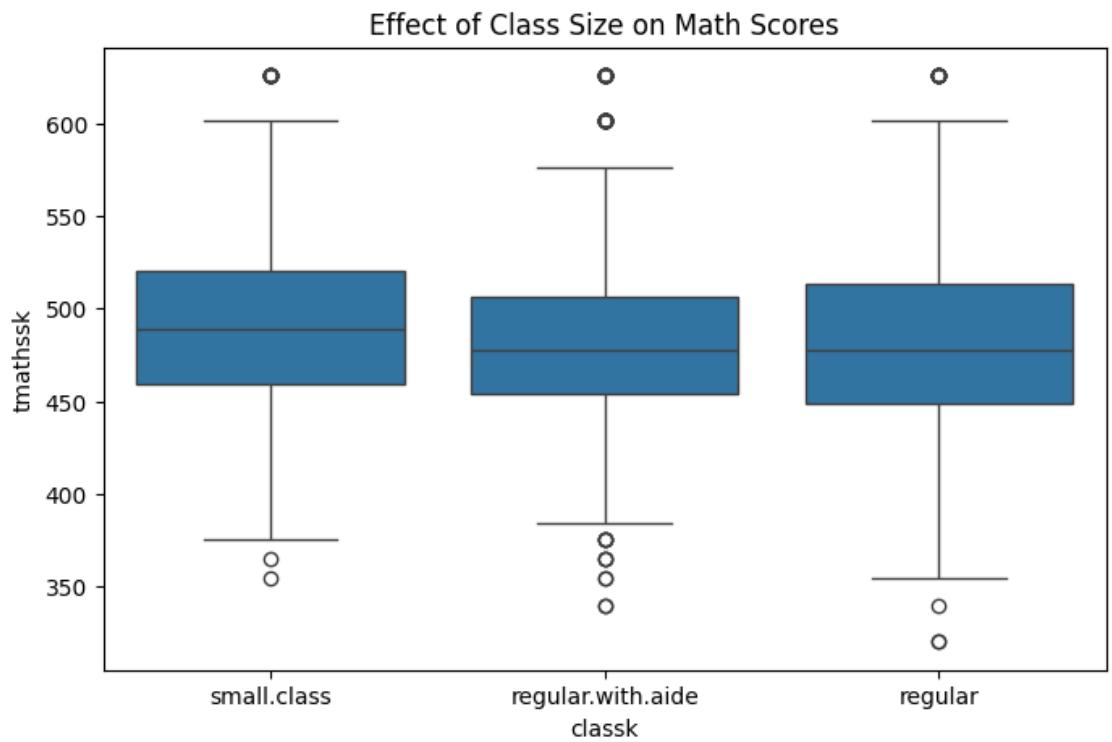
[51]: <Axes: >



[52]:
```
plt.figure(figsize=(10, 6))

plt.subplot(1, 2, 1)
sns.histplot(df['tmathssk'], kde=True)
plt.title("Raw Math Score Distribution")

plt.subplot(1, 2, 2)
sns.histplot(np.log(df['tmathssk']), kde=True)
```

```
plt.title("Log Math Score Distribution")
plt.show()

plt.figure(figsize=(8, 5))
sns.boxplot(x='classk', y='tmathssk', data=df)
plt.title("Effect of Class Size on Math Scores")
plt.xlabel("classk")
plt.ylabel("tmathssk")
plt.show()

Q1 = df['tmathssk'].quantile(0.25)
Q3 = df['tmathssk'].quantile(0.75)
IQR = Q3 - Q1

outliers = df[(df['tmathssk'] < (Q1 - 1.5 * IQR)) | (df['tmathssk'] > (Q3 + 1.5␣
↪* IQR))]
print(outliers)
```






```
      rownames  tmathssk  treadssk             classk  totexpk   sex freelunk  \
22          48       602       456        small.class     14.0   boy      yes
44          93       602       465        small.class     18.0   boy       no
59         137       626       465        small.class      2.0  girl       no
255        523       602       518            regular     12.0  girl       no
280        566       626       510        small.class     14.0  girl       no
…          …         …         …                  …       …    …        …
5614     11312       339       390  regular.with.aide     11.0   boy      yes
5618     11320       626       467        small.class     18.0   boy       no
5631     11349       602       465            regular     13.0  girl       no
5682     11449       365       391            regular      2.0   boy      yes
5685     11458       602       492  regular.with.aide     13.0  girl       no

       race  schidkn
22    white       59
44    white       41
59    black       30
255   white       65
280   white       74
…     …      …
5614  black       45
5618  white       41
5631  white       55
```

```
5682  white       76
5685  white       80

[206 rows x 9 columns]
```

[53]:
```python
import statsmodels.formula.api as smf

formula_1 = 'tmathssk ~ classk + totexpk + race + sex'
model_1 = smf.ols(formula=formula_1, data=df).fit(cov_type='HC1')
print(model_1.summary())
```

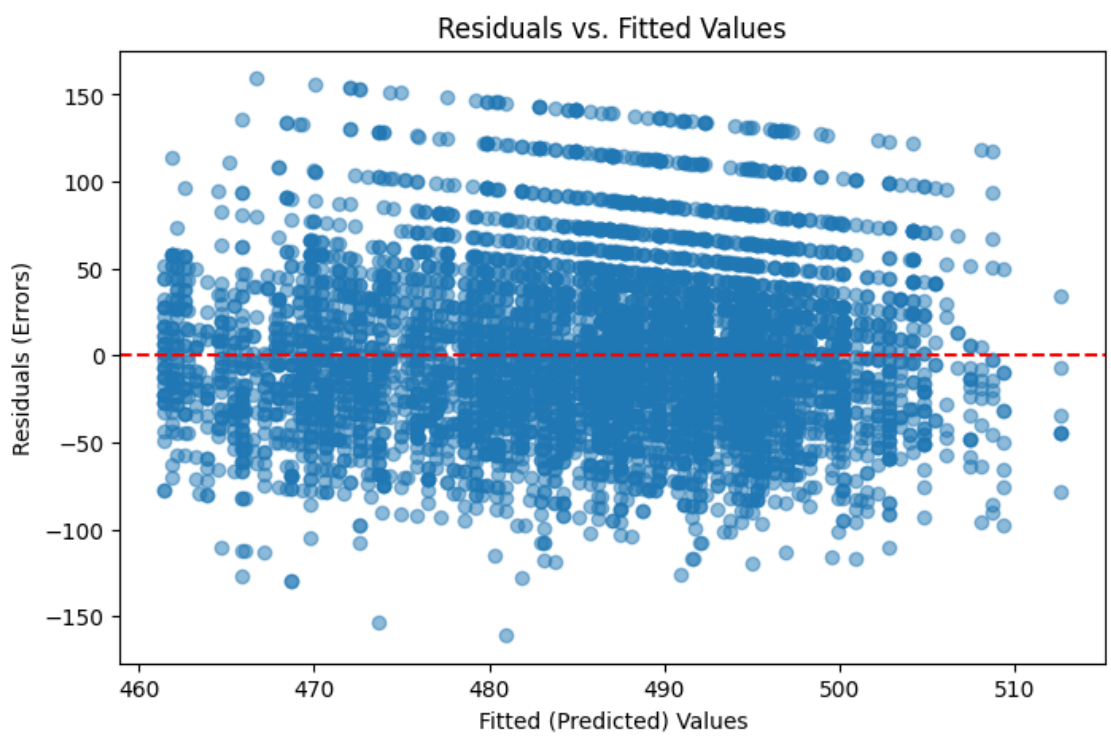
```
                            OLS Regression Results
==============================================================================
Dep. Variable:               tmathssk   R-squared:                       0.050
Model:                            OLS   Adj. R-squared:                  0.049
Method:                 Least Squares   F-statistic:                     53.99
Date:                Tue, 31 Mar 2026   Prob (F-statistic):           4.46e-65
Time:                        19:19:58   Log-Likelihood:                -30232.
No. Observations:                5748   AIC:                         6.048e+04
Df Residuals:                    5741   BIC:                         6.052e+04
Df Model:                           6
Covariance Type:                  HC1
================================================================================
===============
                                 coef    std err          z      P>|z|
[0.025      0.975]
--------------------------------------------------------------------------------
---------------
Intercept                    461.9061      1.697    272.116      0.000
458.579     465.233
classk[T.regular.with.aide]    -0.4498      1.437     -0.313      0.754
-3.267       2.367
classk[T.small.class]           8.1669      1.568      5.208      0.000
5.093      11.240
race[T.other]                   5.3981      9.881      0.546      0.585
-13.968      24.764
race[T.white]                  17.1083      1.351     12.661      0.000
14.460      19.757
sex[T.girl]                     7.8316      1.229      6.372      0.000
5.423      10.241
totexpk                         0.6543      0.113      5.781      0.000
0.432       0.876
==============================================================================
Omnibus:                      245.312   Durbin-Watson:                   2.001
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              279.216
Skew:                           0.510   Prob(JB):                     2.34e-61
Kurtosis:                       3.353   Cond. No.                         162.
==============================================================================
```

```
Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
```

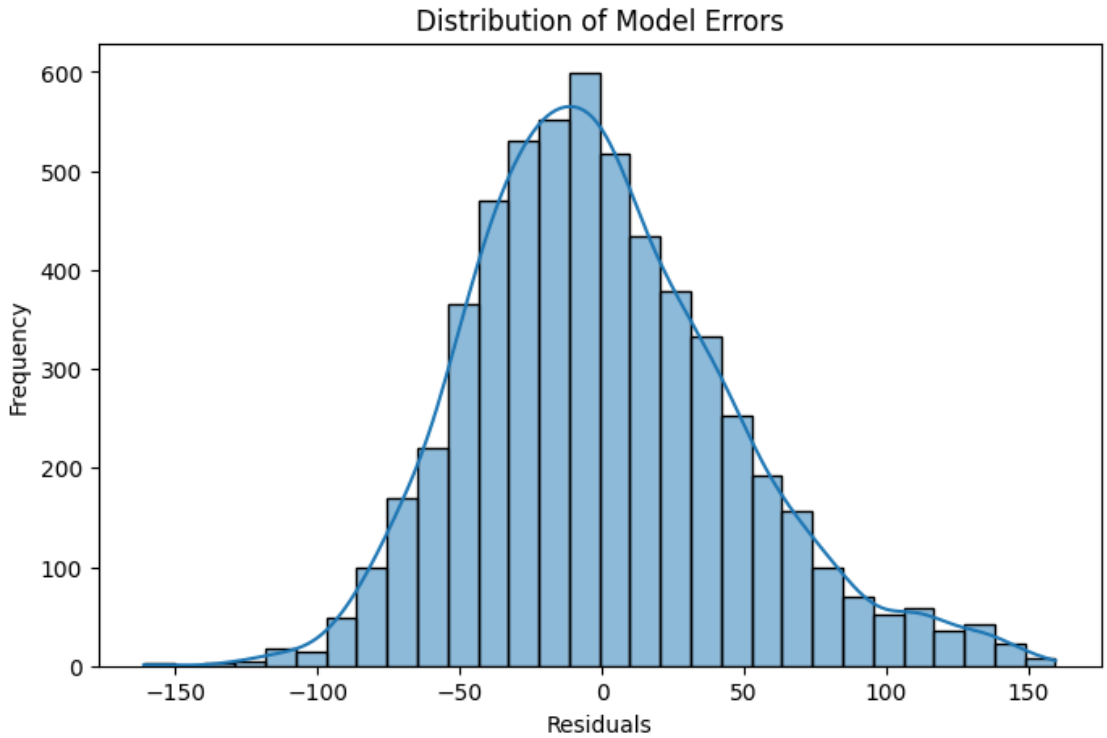
[54]:
```python
formula_interact = 'tmathssk ~ classk * totexpk'
model_interact = smf.ols(formula=formula_interact, data=df).fit(cov_type='HC1')
print(model_interact.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:               tmathssk   R-squared:                       0.018
Model:                            OLS   Adj. R-squared:                  0.018
Method:                 Least Squares   F-statistic:                     22.07
Date:                Tue, 31 Mar 2026   Prob (F-statistic):           5.72e-22
Time:                        19:19:58   Log-Likelihood:                -30326.
No. Observations:                5748   AIC:                         6.066e+04
Df Residuals:                    5742   BIC:                         6.070e+04
Df Model:                           5
Covariance Type:                  HC1
==============================================================================================
=======================
                                            coef    std err          z      P>|z|
[0.025      0.975]
----------------------------------------------------------------------------------------------
-----------------------
Intercept                              474.6329      2.028    234.029      0.000
470.658     478.608
classk[T.regular.with.aide]             -3.1544      2.795     -1.129      0.259
-8.633       2.324
classk[T.small.class]                   14.3657      3.032      4.738      0.000
8.423      20.308
totexpk                                  0.9504      0.192      4.955      0.000
0.574       1.326
classk[T.regular.with.aide]:totexpk      0.2253      0.262      0.860      0.390
-0.288       0.739
classk[T.small.class]:totexpk           -0.6755      0.282     -2.395      0.017
-1.228      -0.123
==============================================================================
Omnibus:                      195.967   Durbin-Watson:                   1.981
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              216.629
Skew:                           0.456   Prob(JB):                     9.11e-48
Kurtosis:                       3.267   Cond. No.                         76.5
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
```

[55]:
```
import matplotlib.pyplot as plt
import seaborn as sns

# Extract the fitted values and residuals from your model
fitted_vals = model_1.fittedvalues
residuals = model_1.resid

# Create the scatter plot
plt.figure(figsize=(8, 5))
plt.scatter(fitted_vals, residuals, alpha=0.5)
plt.axhline(y=0, color='r', linestyle='--') # Adds a red line at zero
plt.xlabel("Fitted (Predicted) Values")
plt.ylabel("Residuals (Errors)")
plt.title("Residuals vs. Fitted Values")
plt.show()
```



[56]:
```
# Create a histogram of the residuals
plt.figure(figsize=(8, 5))
sns.histplot(residuals, kde=True, bins=30)
plt.xlabel("Residuals")
plt.ylabel("Frequency")
plt.title("Distribution of Model Errors")
plt.show()
```



[57]:
```
print(df.groupby('classk')[['tmathssk', 'totexpk']].describe())
```

```
                  tmathssk                                                          \
                     count        mean        std    min    25%    50%    75%
classk
regular             2000.0  483.261000  47.500664  320.0  449.0  478.0  513.0
regular.with.aide   2015.0  483.009926  46.111625  339.0  454.0  478.0  506.0
small.class         1733.0  491.470283  49.468723  354.0  459.0  489.0  520.0

                        totexpk                                                 \
                    max   count      mean       std  min  25%   50%   75%
classk
regular           626.0  2000.0  9.078000  5.724448  0.0  4.0   9.0  13.0
regular.with.aide 626.0  2015.0  9.807444  5.810961  0.0  5.0  10.0  13.0
small.class       626.0  1733.0  8.990767  5.730971  0.0  4.0   8.0  13.0


                    max
classk
regular            24.0
regular.with.aide  27.0
small.class        27.0
```

[58]:
```
mod1 = smf.ols('tmathssk ~ classk', data=df).fit(cov_type='HC1')
print(mod1.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:               tmathssk   R-squared:                       0.006
Model:                            OLS   Adj. R-squared:                  0.006
Method:                 Least Squares   F-statistic:                     17.78
Date:                Tue, 31 Mar 2026   Prob (F-statistic):           2.01e-08
Time:                        19:19:58   Log-Likelihood:                -30361.
No. Observations:                5748   AIC:                         6.073e+04
Df Residuals:                    5745   BIC:                         6.075e+04
Df Model:                           2
Covariance Type:                  HC1
==================================================================================
===============
                                 coef    std err          z      P>|z|
[0.025      0.975]
----------------------------------------------------------------------------------
---------------
Intercept                     483.2610      1.062    454.980      0.000
481.179     485.343
classk[T.regular.with.aide]    -0.2511      1.478     -0.170      0.865
-3.147       2.645
classk[T.small.class]           8.2093      1.594      5.151      0.000
5.085      11.333
==============================================================================
Omnibus:                      204.091   Durbin-Watson:                   1.972
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              226.743
Skew:                           0.465   Prob(JB):                     5.80e-50
Kurtosis:                       3.284   Cond. No.                         3.67
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
```

[59]:
```
mod2 = smf.ols('tmathssk ~ classk + totexpk + race + sex', data=df).
 ↪fit(cov_type='HC1')
print(mod2.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:               tmathssk   R-squared:                       0.050
Model:                            OLS   Adj. R-squared:                  0.049
Method:                 Least Squares   F-statistic:                     53.99
Date:                Tue, 31 Mar 2026   Prob (F-statistic):           4.46e-65
Time:                        19:19:58   Log-Likelihood:                -30232.
No. Observations:                5748   AIC:                         6.048e+04
```

```
Df Residuals:                     5741   BIC:                         6.052e+04
Df Model:                            6
Covariance Type:                   HC1
==============================================================================
===============
                                 coef    std err          z      P>|z|
[0.025      0.975]
------------------------------------------------------------------------------
---------------
Intercept                     461.9061      1.697    272.116      0.000
458.579     465.233
classk[T.regular.with.aide]    -0.4498      1.437     -0.313      0.754
-3.267       2.367
classk[T.small.class]           8.1669      1.568      5.208      0.000
5.093      11.240
race[T.other]                   5.3981      9.881      0.546      0.585
-13.968      24.764
race[T.white]                  17.1083      1.351     12.661      0.000
14.460      19.757
sex[T.girl]                     7.8316      1.229      6.372      0.000
5.423      10.241
totexpk                         0.6543      0.113      5.781      0.000
0.432       0.876
==============================================================================
Omnibus:                      245.312   Durbin-Watson:                   2.001
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              279.216
Skew:                           0.510   Prob(JB):                     2.34e-61
Kurtosis:                       3.353   Cond. No.                         162.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
```

[60]:
```python
mod3 = smf.ols('tmathssk ~ classk * totexpk + race + sex', data=df).
 ↪fit(cov_type='HC1')
print(mod3.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:               tmathssk   R-squared:                       0.053
Model:                            OLS   Adj. R-squared:                  0.052
Method:                 Least Squares   F-statistic:                     45.14
Date:                Tue, 31 Mar 2026   Prob (F-statistic):           7.55e-71
Time:                        19:19:58   Log-Likelihood:                -30224.
No. Observations:                5748   AIC:                         6.047e+04
Df Residuals:                    5739   BIC:                         6.053e+04
Df Model:                           8
Covariance Type:                   HC1
```

```
==============================================================================
========================
                                          coef    std err          z      P>|z|
[0.025      0.975]
------------------------------------------------------------------------------
------------------------
Intercept                              460.4235      2.182    210.972      0.000
456.146     464.701
classk[T.regular.with.aide]             -3.0226      2.762     -1.094      0.274
-8.436       2.390
classk[T.small.class]                   15.3169      2.999      5.108      0.000
9.439      21.194
race[T.other]                            6.1474      9.865      0.623      0.533
-13.187      25.482
race[T.white]                           17.3620      1.338     12.976      0.000
14.740      19.984
sex[T.girl]                              7.8125      1.228      6.362      0.000
5.406      10.219
totexpk                                  0.7993      0.195      4.099      0.000
0.417       1.181
classk[T.regular.with.aide]:totexpk      0.2521      0.264      0.953      0.340
-0.266       0.770
classk[T.small.class]:totexpk           -0.7941      0.281     -2.827      0.005
-1.345      -0.244
==============================================================================
Omnibus:                      235.513   Durbin-Watson:                   2.001
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              266.484
Skew:                           0.500   Prob(JB):                     1.36e-58
Kurtosis:                       3.335   Cond. No.                         181.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
```